

CardamomOT — analysis report

/Users/eventre/Documents/GitHub/CardamomOT/experimental_datasets/Kameneva

Generated on 2026-10-05 18:33

Run

Split	train
Stimulus penalisation	0.2
Prior penalisation	1.0
Genes	97
Stimuli	1
Observed cells	1260
Timepoints	0, 8, 32, 48, 64, 80
Cell types	precursors, sympathoblasts, intermediate, chromaffin
Network shown	inter_simul.npy, network #0 / 1
UMAP preprocessing	normalise=False, log1p=True

Perturbations (perturbation_simulation)

✓	P1 KO CHGA (50%)
✓	P2 KO CHGA
✓	P3 OV STMN2
✓	P4 KO CHGA · OV STMN2

Contents

1. Generative model — UMAPs of data, trajectories, NB mixture, network modes and simulation; cell-type proportions; gene-pair correlations; proteins
2. Gene regulatory network — regulatory power (violin plots) and top-10 regulators
3. In-silico perturbations — overview across KO/OV, then one page per perturbation
4. Proliferation — prior vs learned net rates, population growth, proteins driving growth
5. Learned dynamics — mRNA and protein velocity fields (mechanistic and along trajectories), summary on mRNA
6. Held-out test cells — predictions with the network fixed vs the test data

0. Per-cell sequencing depth

Diagnostic on the whole transcriptome within homogeneous groups (sample, time, cell type); counts are modelled as $NB(k, c / s_i)$ when the depth factor s_i is applied.

Depth

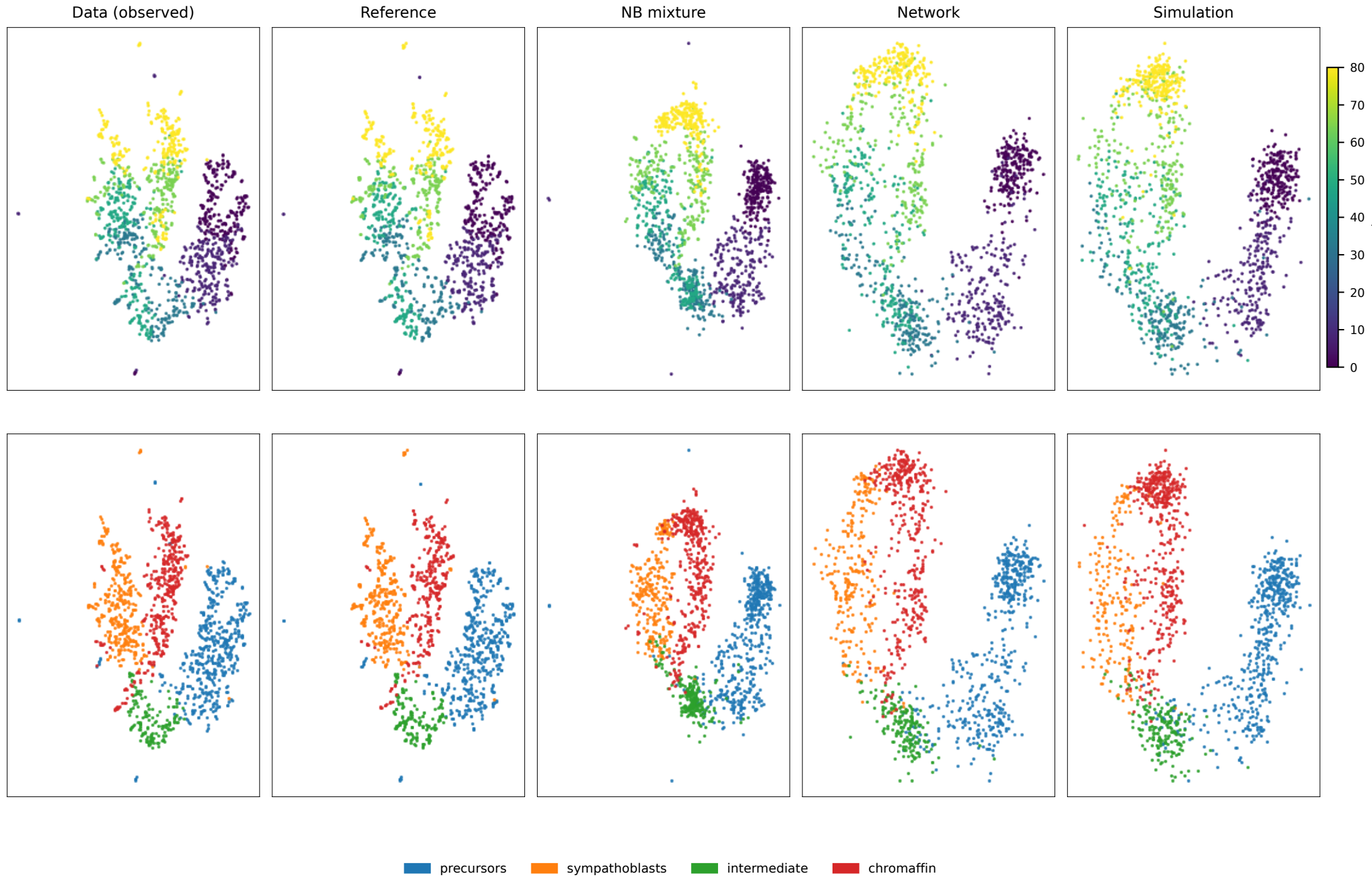
not estimated

Reason

97 genes < 10000

1. Generative model — trajectories and simulation

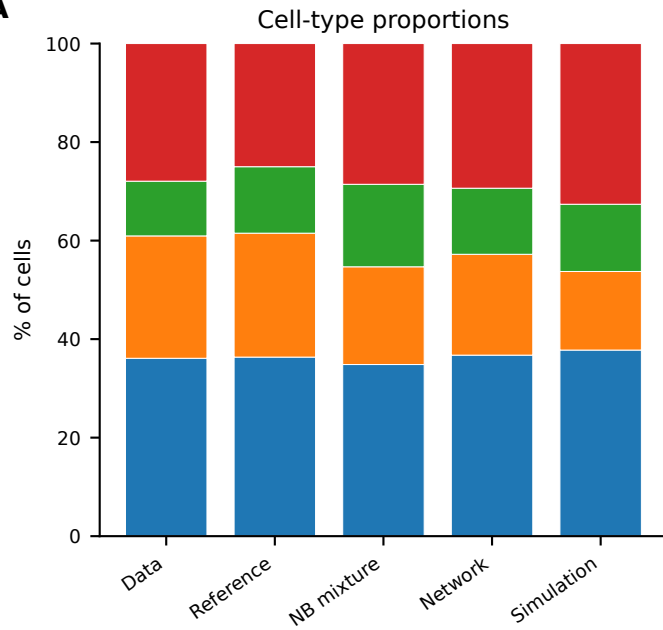
Joint UMAP of the observed data, the trajectories (Reference: one state per ancestor, i.e. without proliferation, as the simulation), NB mixture and network-driven modes along them, and the full simulation.



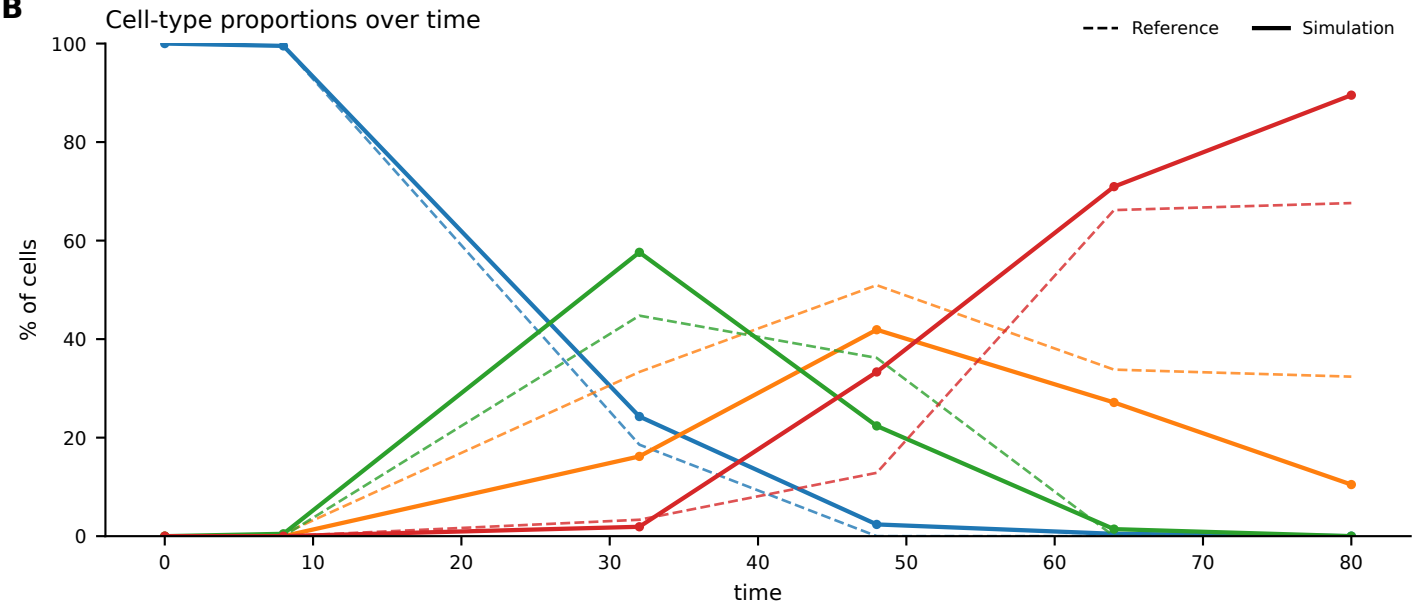
1. Generative model — quantitative checks

Reference = RNA of the trajectories equivalent to the simulation (without proliferation).

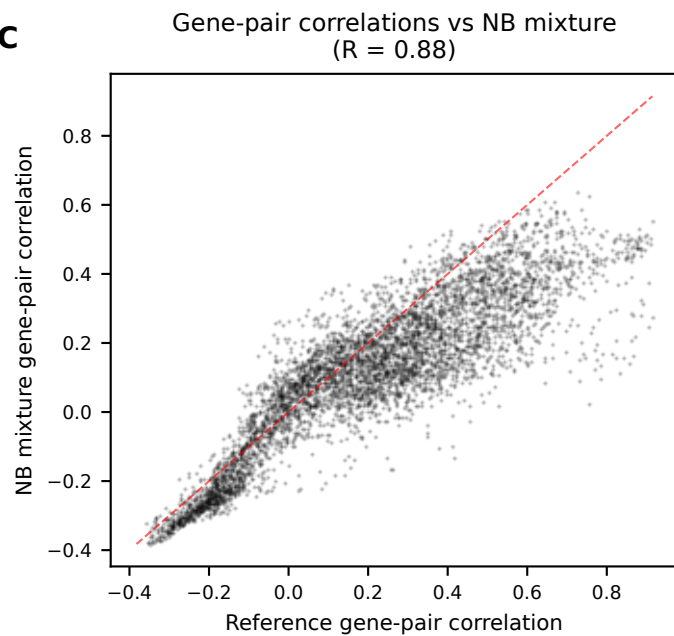
A



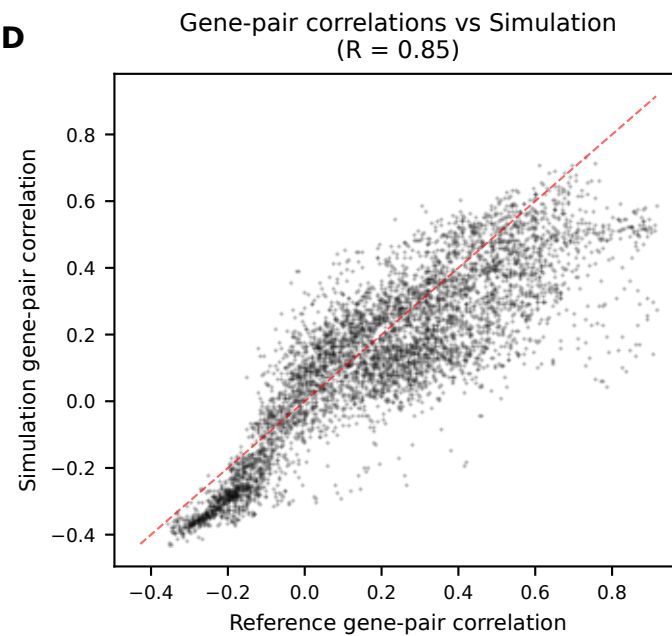
B



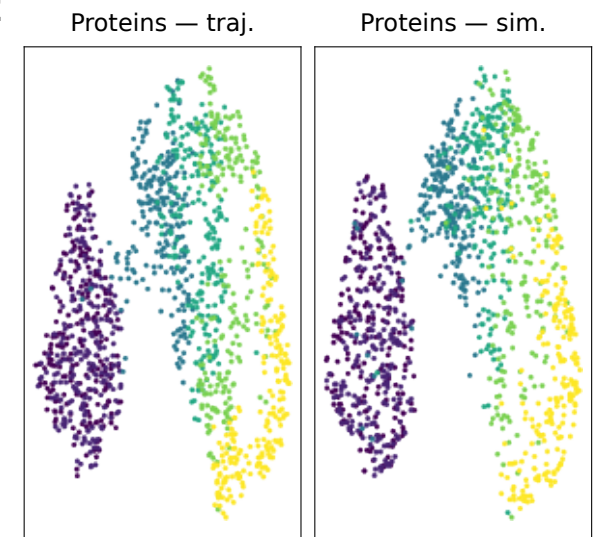
C



D



E



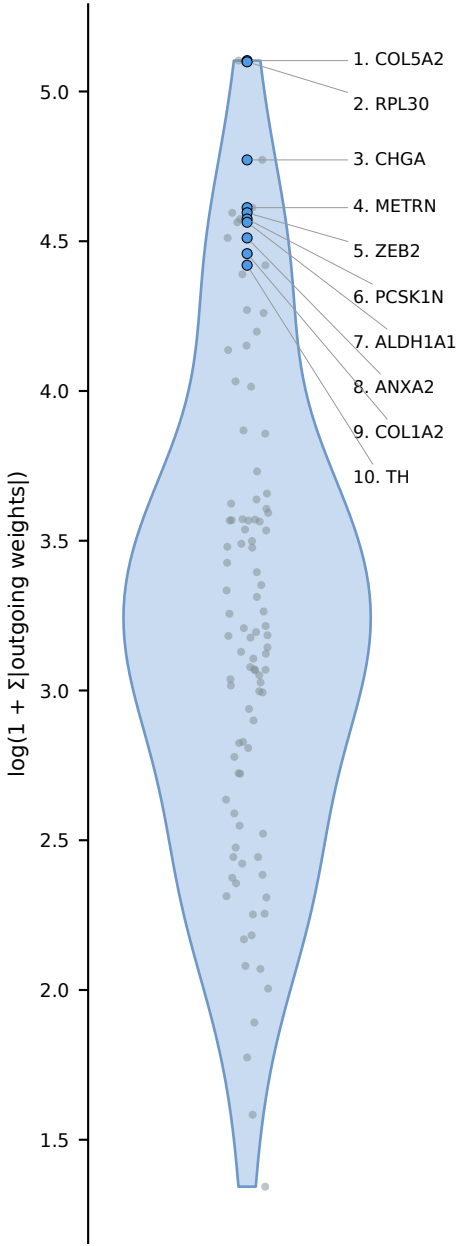
precursors sympathoblasts intermediate chromaffin

2. Gene regulatory network — regulatory power

Self-interactions excluded. Top 10 genes labelled; stimulus not shown (stimulus = 0.2 < 0.5).

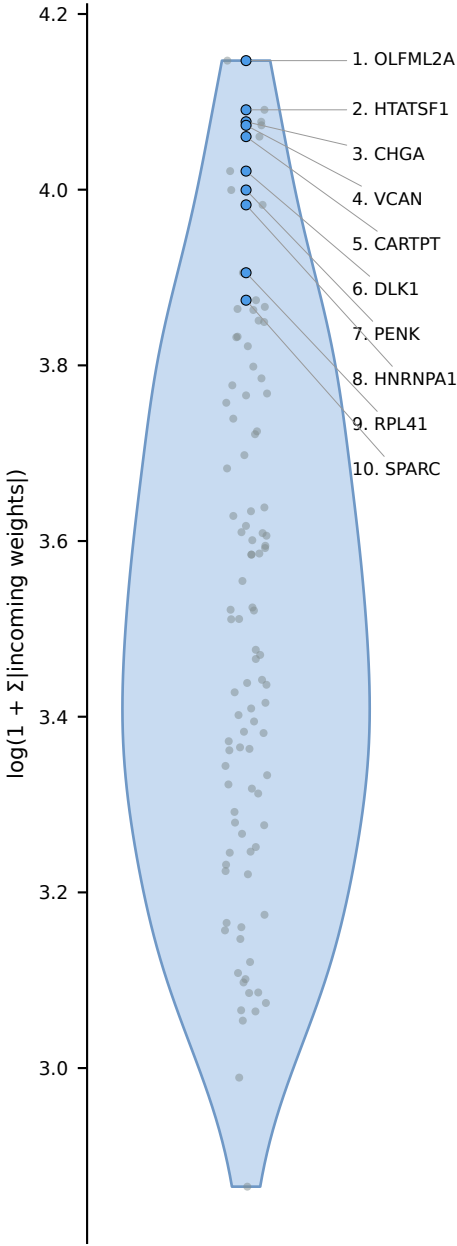
A

Outgoing regulation (regulators)



B

Incoming regulation (targets)



C

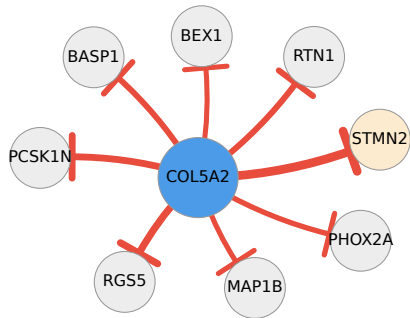
Top regulators

#	Regulator	$\sum w $	# act.	# inh.
1	COL5A2	163.52	1	42
2	RPL30	162.86	73	14
3	CHGA	117.10	53	15
4	METRN	99.70	25	19
5	ZEB2	97.99	2	43
6	PCSK1N	95.95	42	13
7	ALDH1A1	94.90	8	57
8	ANXA2	90.02	37	17
9	COL1A2	85.36	22	14
10	TH	82.08	36	21

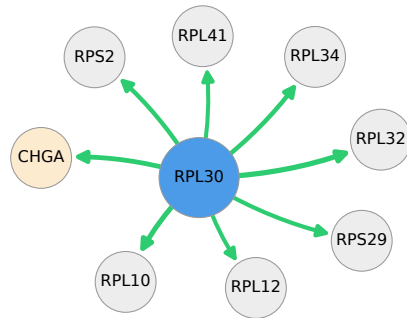
2. Gene regulatory network — top regulators and their main targets

Top 8 targets per regulator; edge width $\propto$ |weight| (global scale). Perturbed genes (KO/OV) are highlighted in beige.

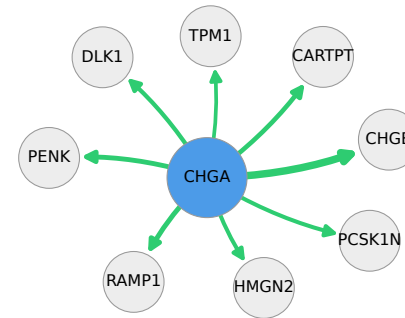
#1 COL5A2



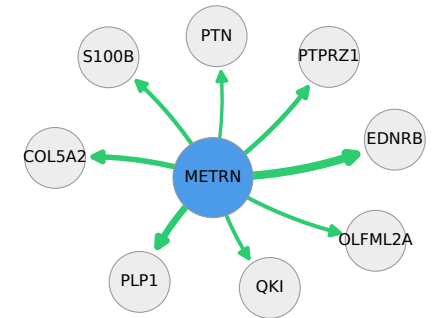
#2 RPL30



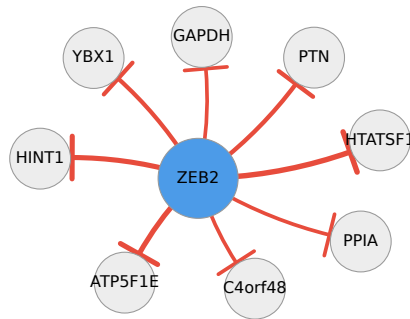
#3 CHGA



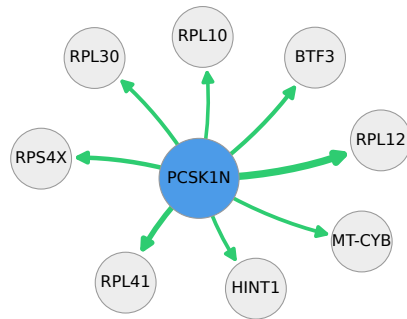
#4 METRN



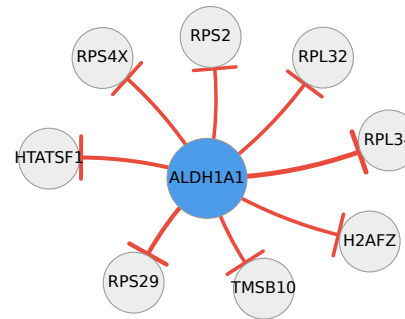
#5 ZEB2



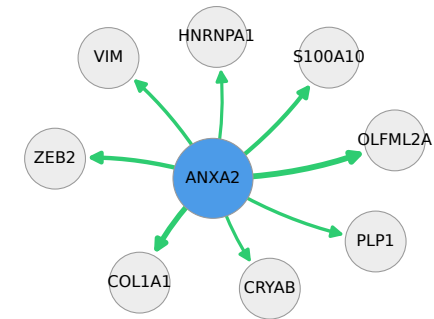
#6 PCSK1N



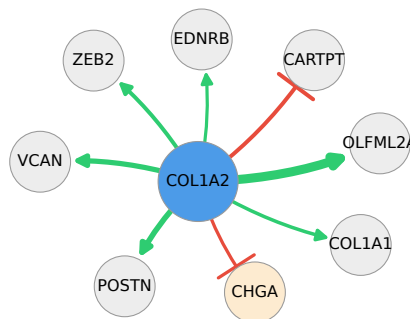
#7 ALDH1A1



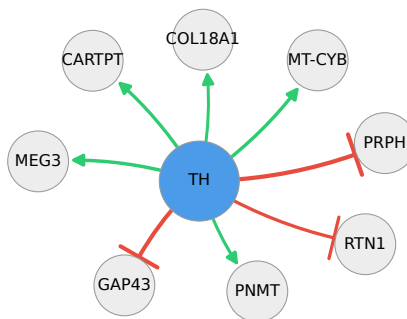
#8 ANXA2



#9 COL1A2



#10 TH



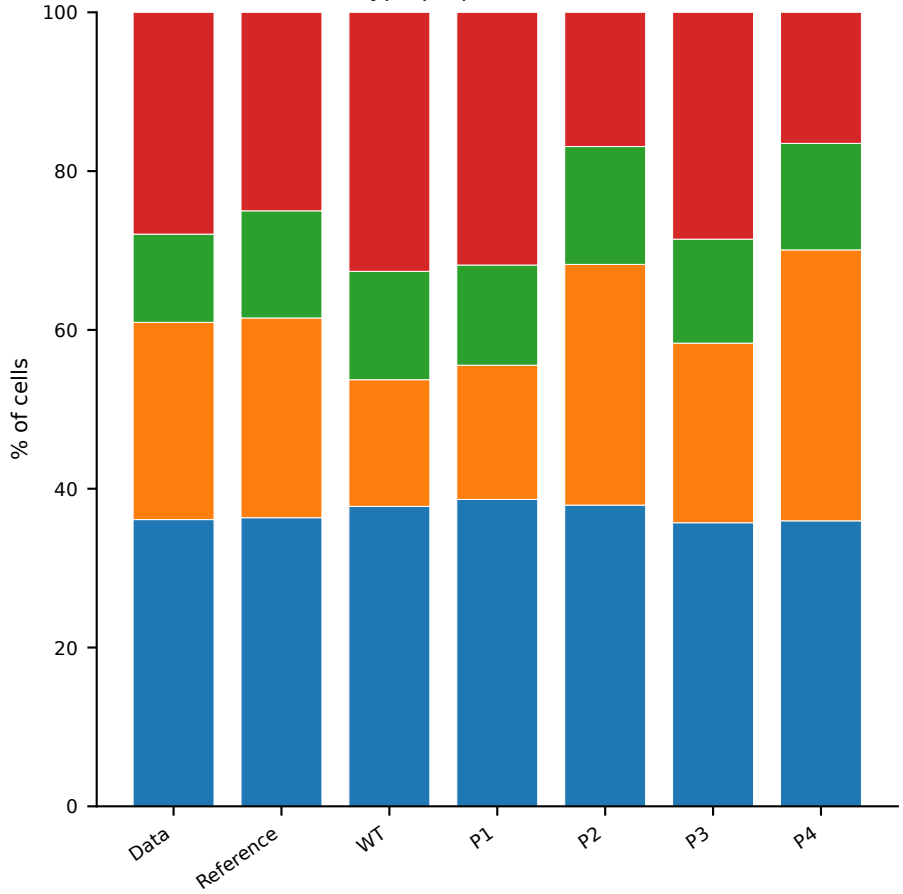
— Activation — Inhibition — Top regulator

3. In-silico perturbations — overview

Cell types predicted by a random forest trained on the observed data.

A

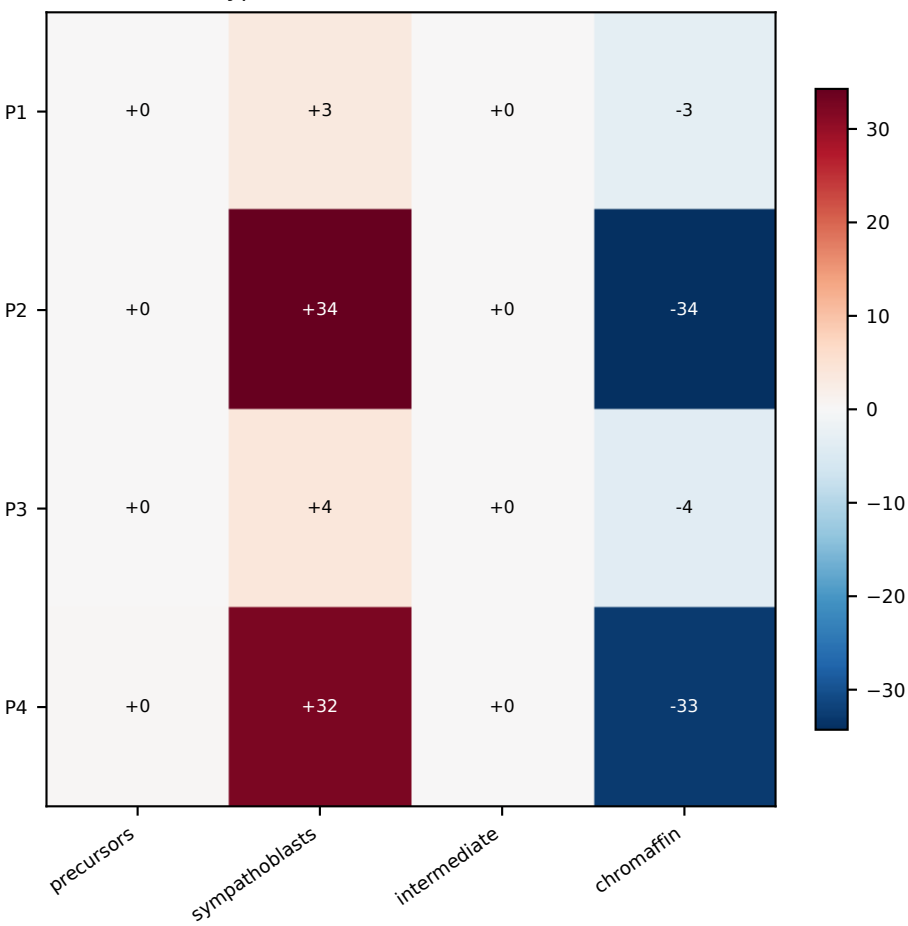
Cell-type proportions (all times)



P1 KO CHGA (50%)
P2 KO CHGA

B

Cell-type shift vs WT simulation at t = 80

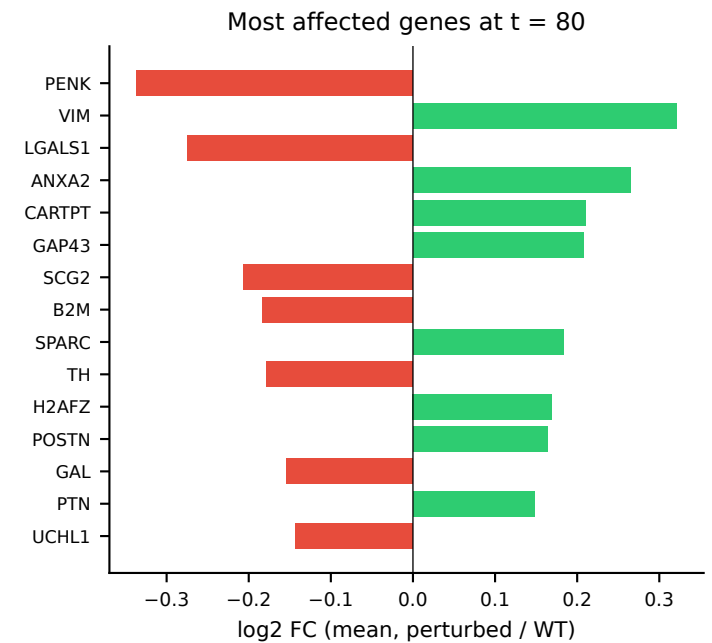
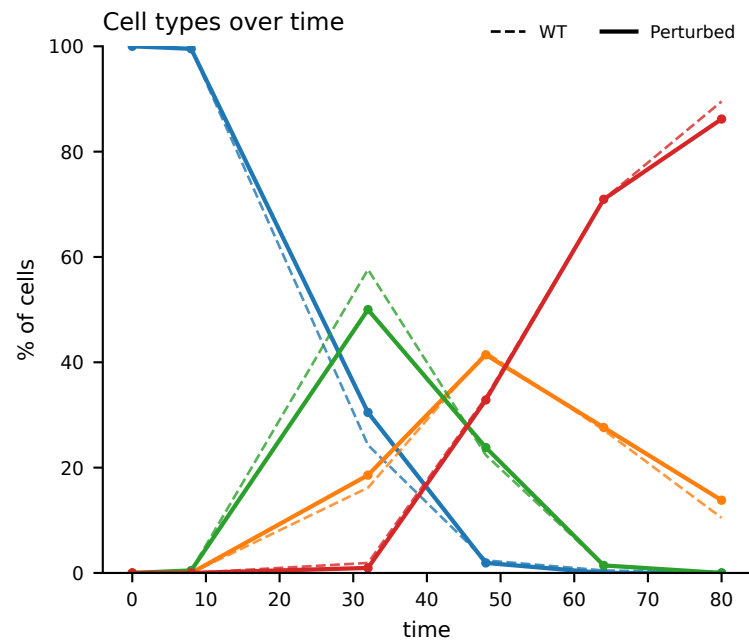
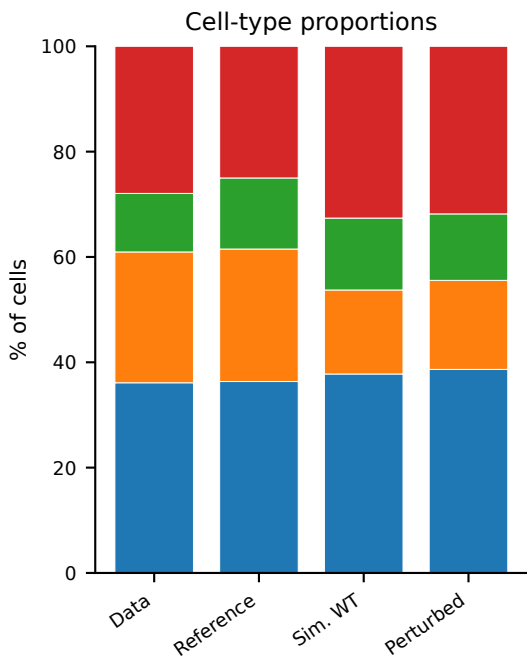
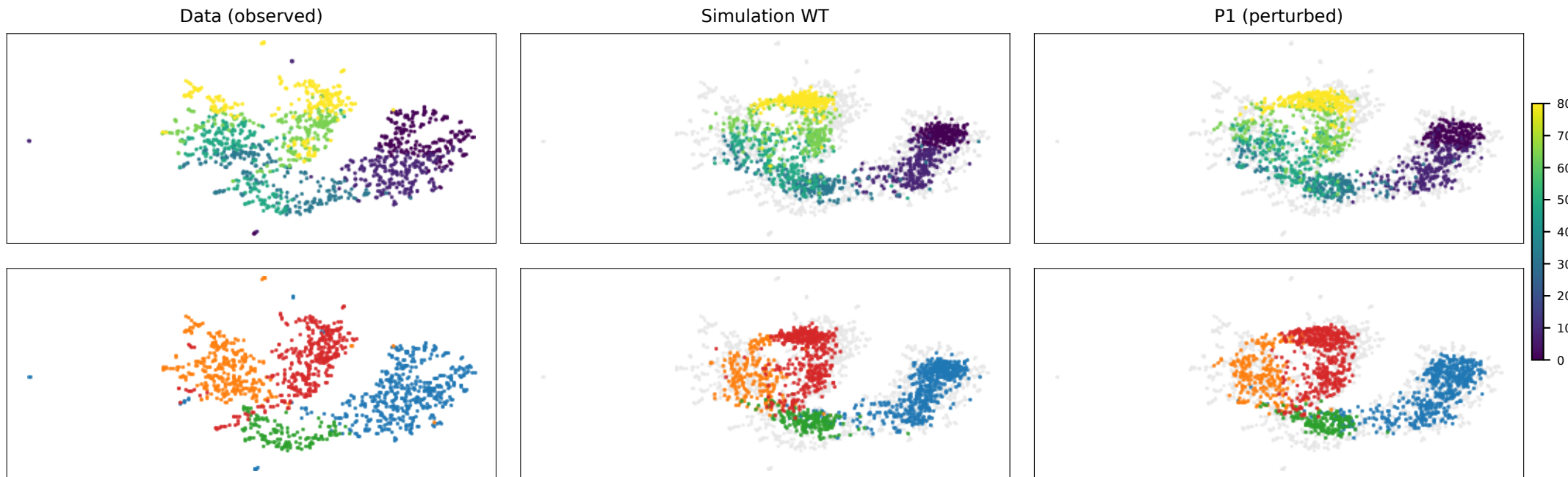


P3 OV STMN2
P4 KO CHGA · OV STMN2

precursors sympathoblasts intermediate chromaffin

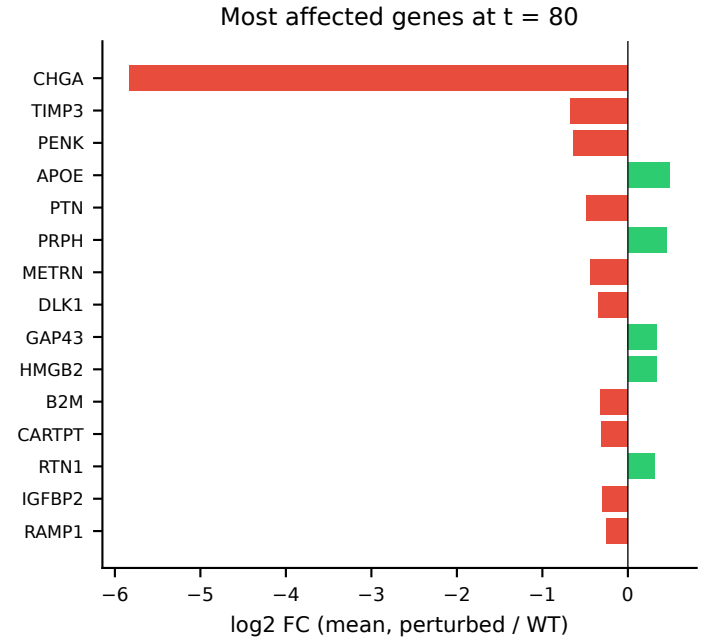
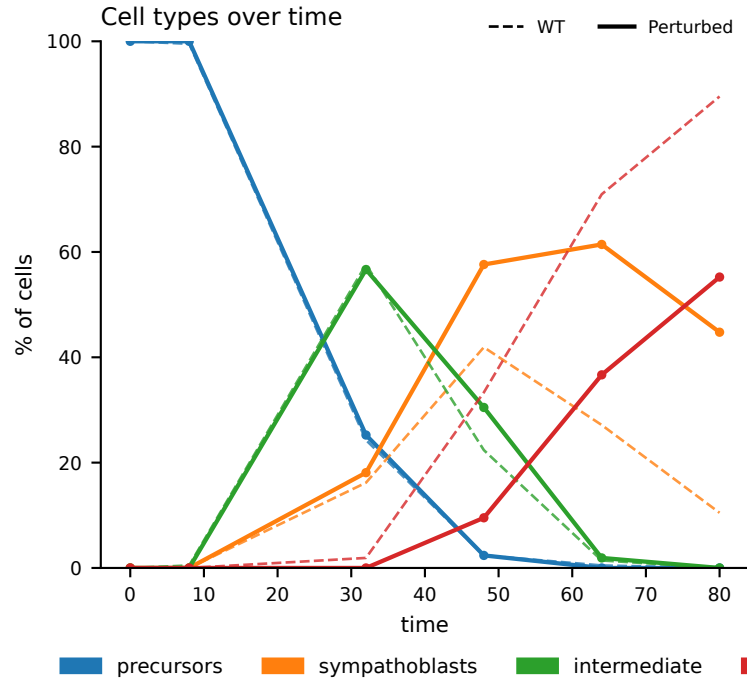
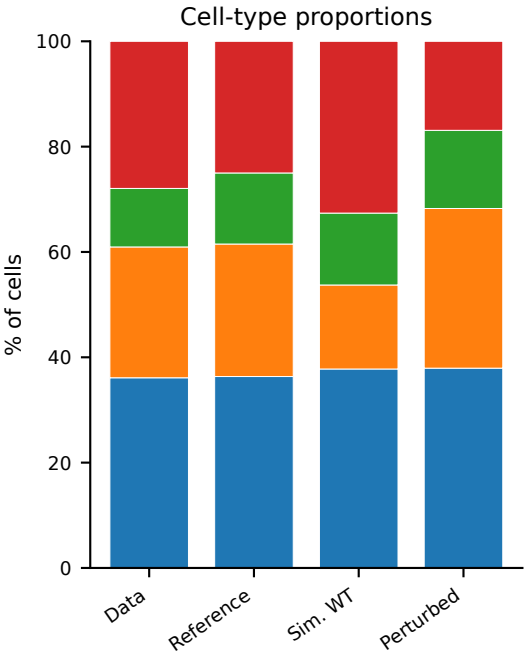
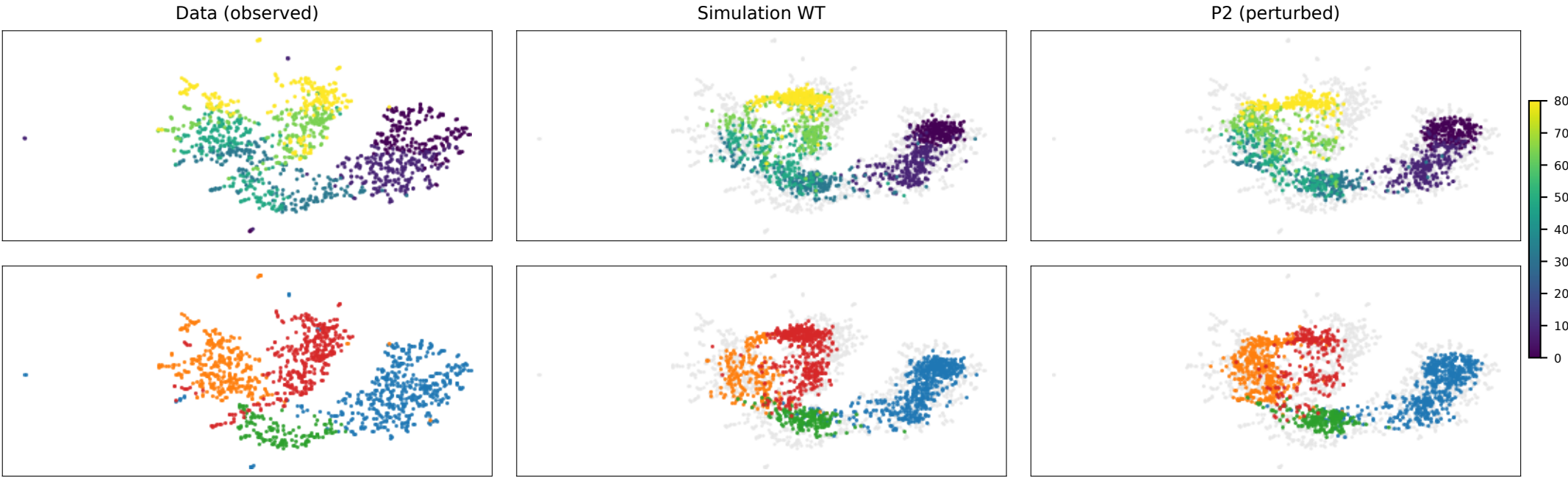
3. Perturbation P1 — KO CHGA (50%)

KO_CHGApct50_OV_none. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



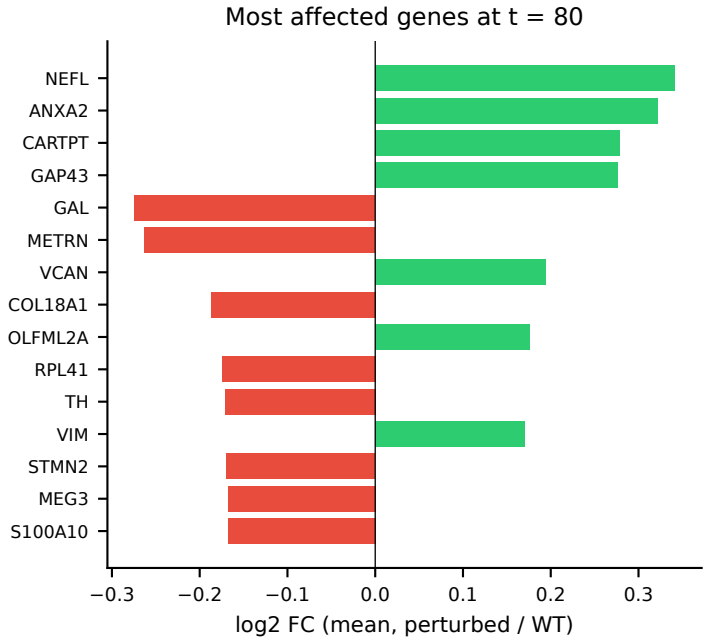
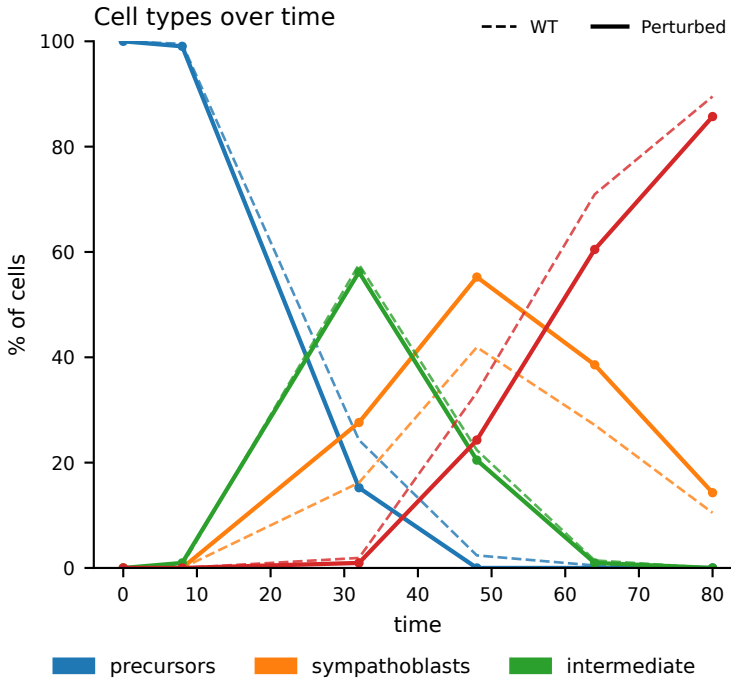
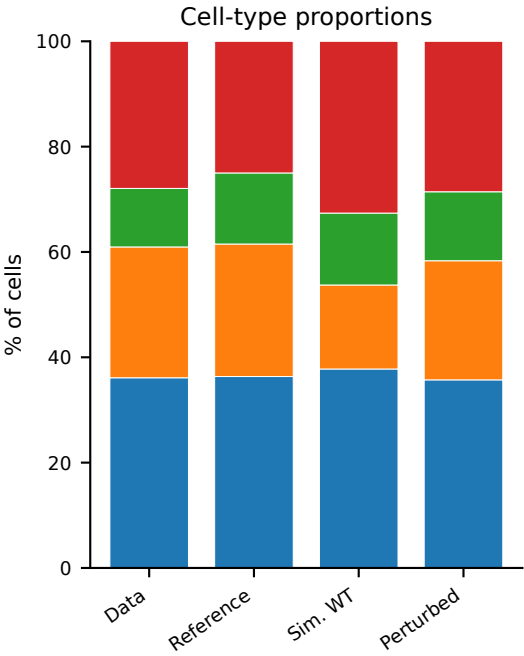
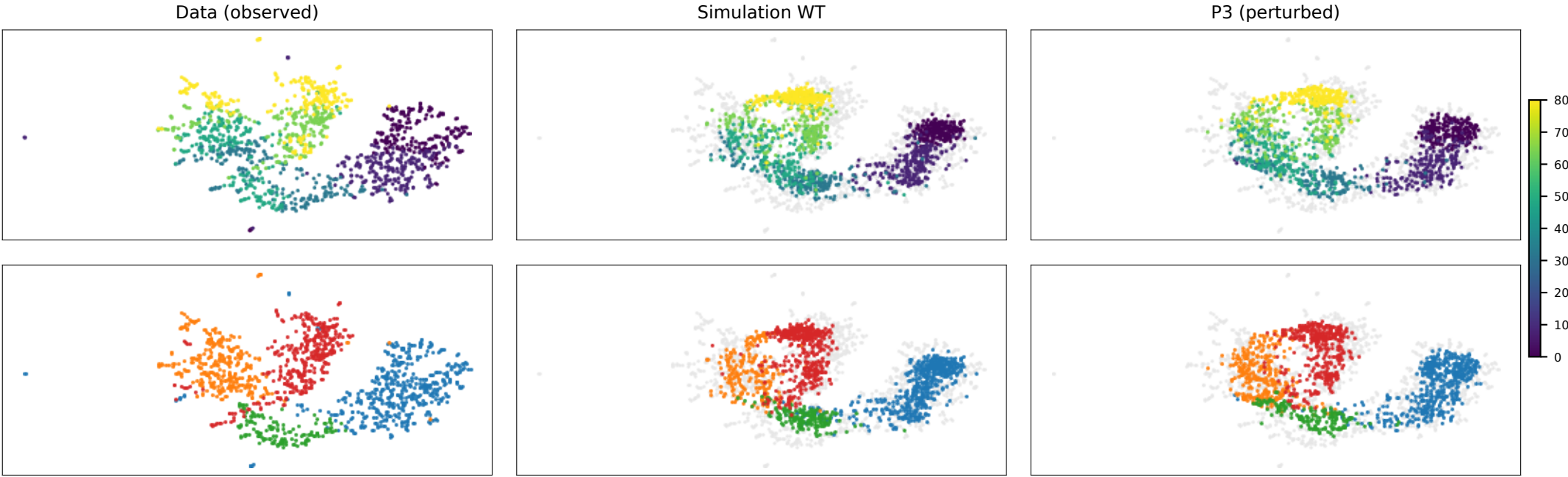
3. Perturbation P2 — KO CHGA

KO_CHGA_OV_none. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



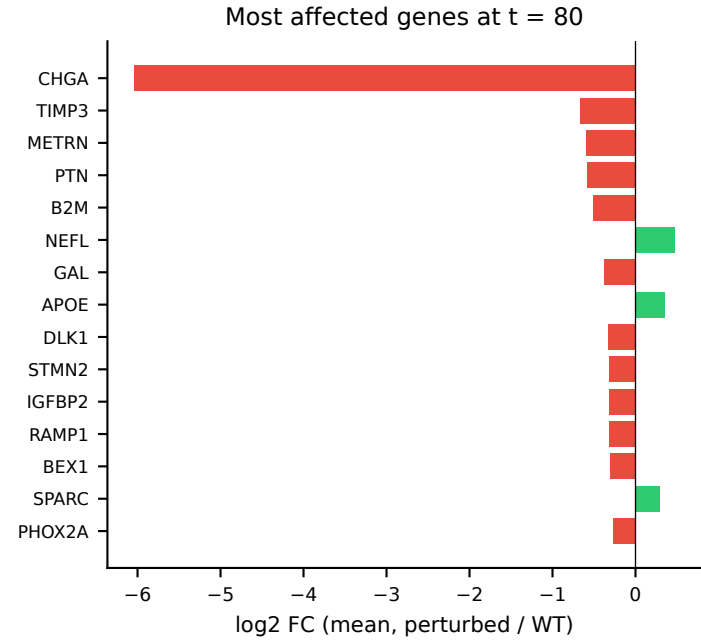
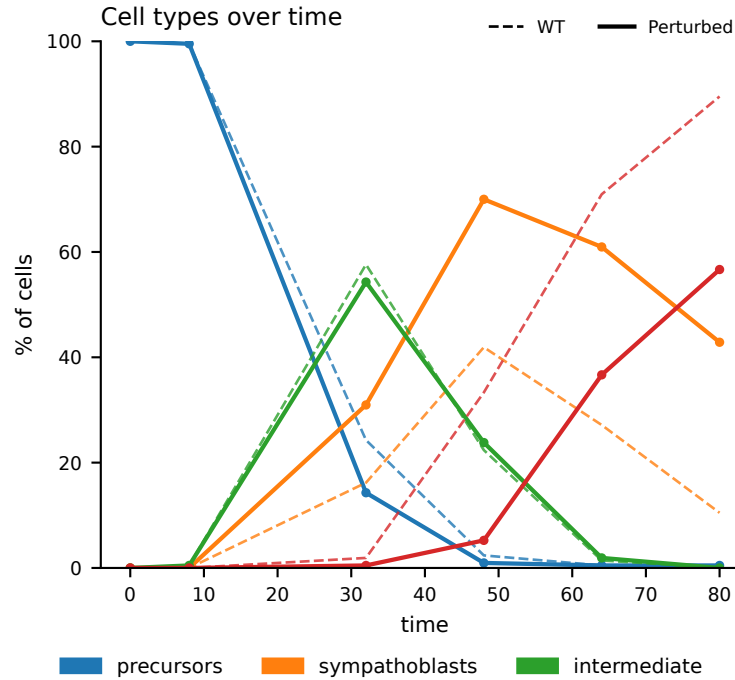
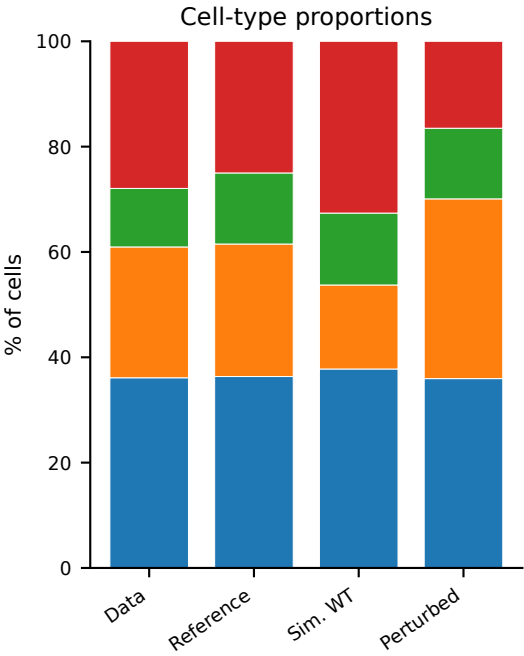
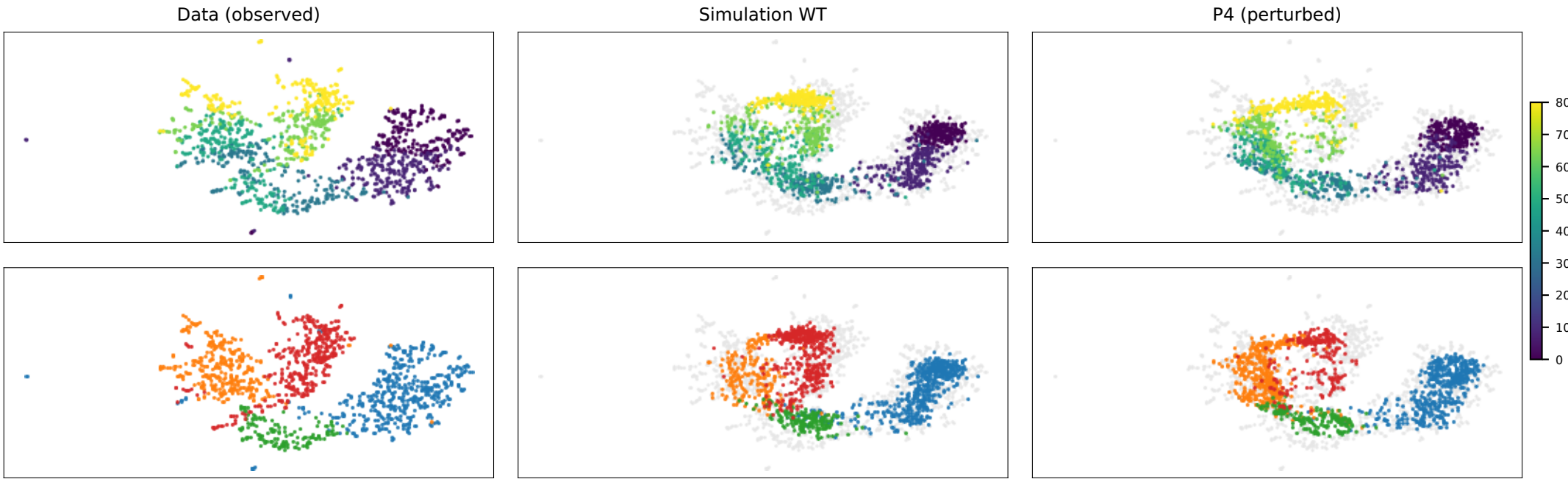
3. Perturbation P3 — OV STMN2

KO_none_OV_STMN2. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



3. Perturbation P4 — KO CHGA · OV STMN2

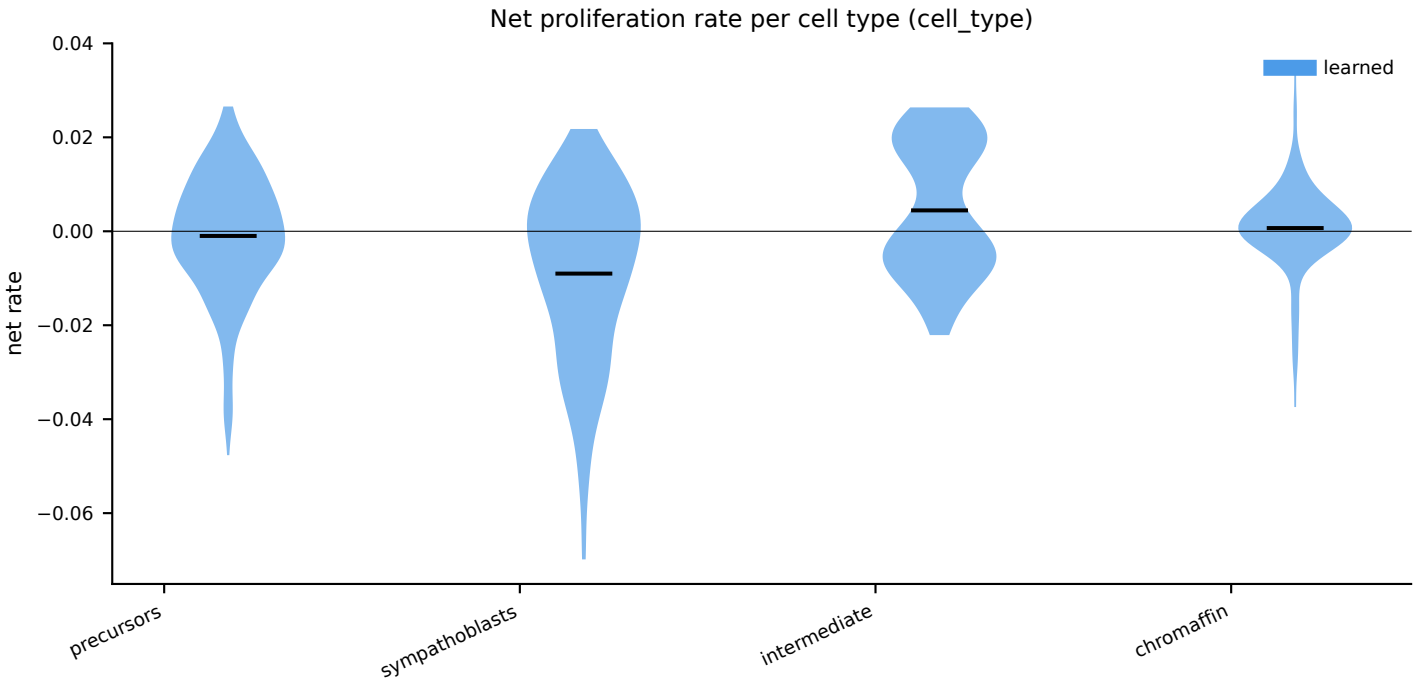
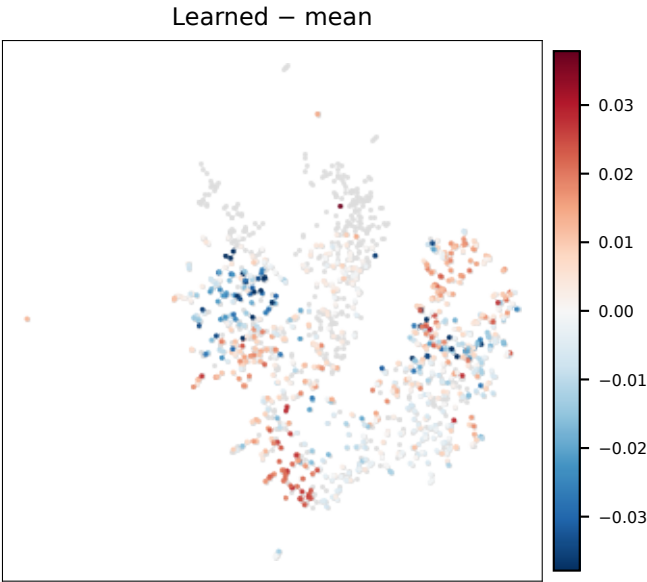
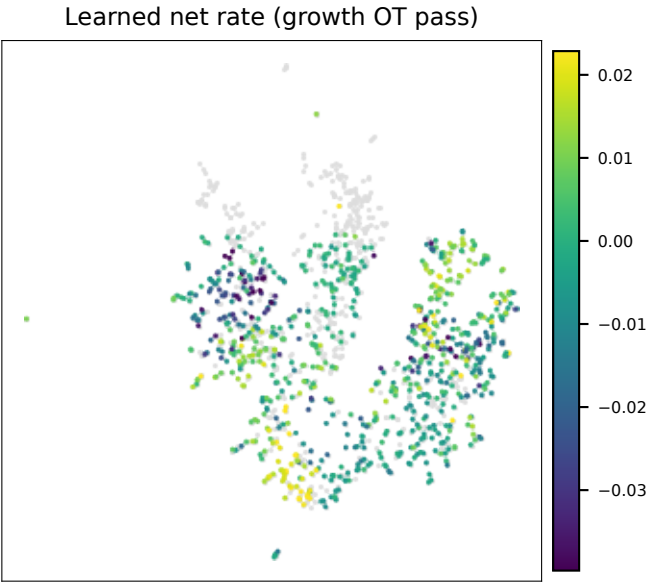
KO_CHGA_OV_STMN2. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



4. Proliferation — prior vs learned net rate (birth – death)

Learned = growth OT pass, mean over the trajectory states behind each observed cell (grey = cell not reached). Rates per time unit of the data.

No prior provided:
uniform net rate
(only relative rates are learned)

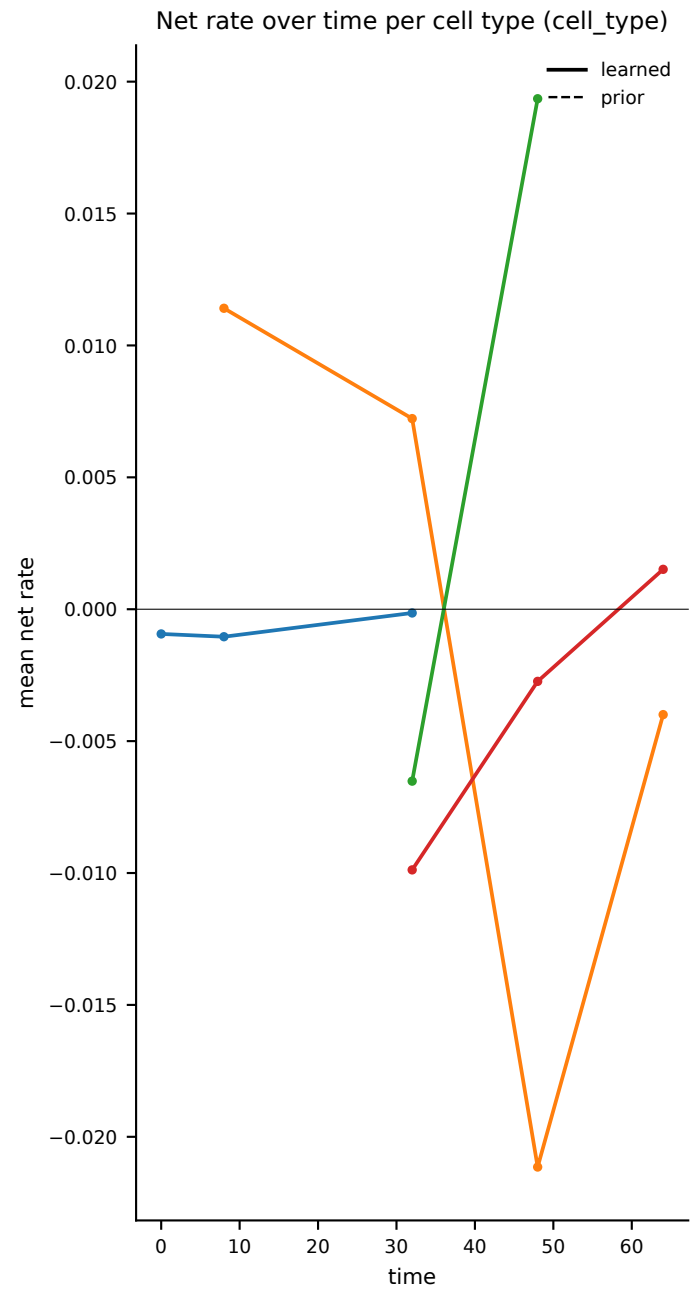
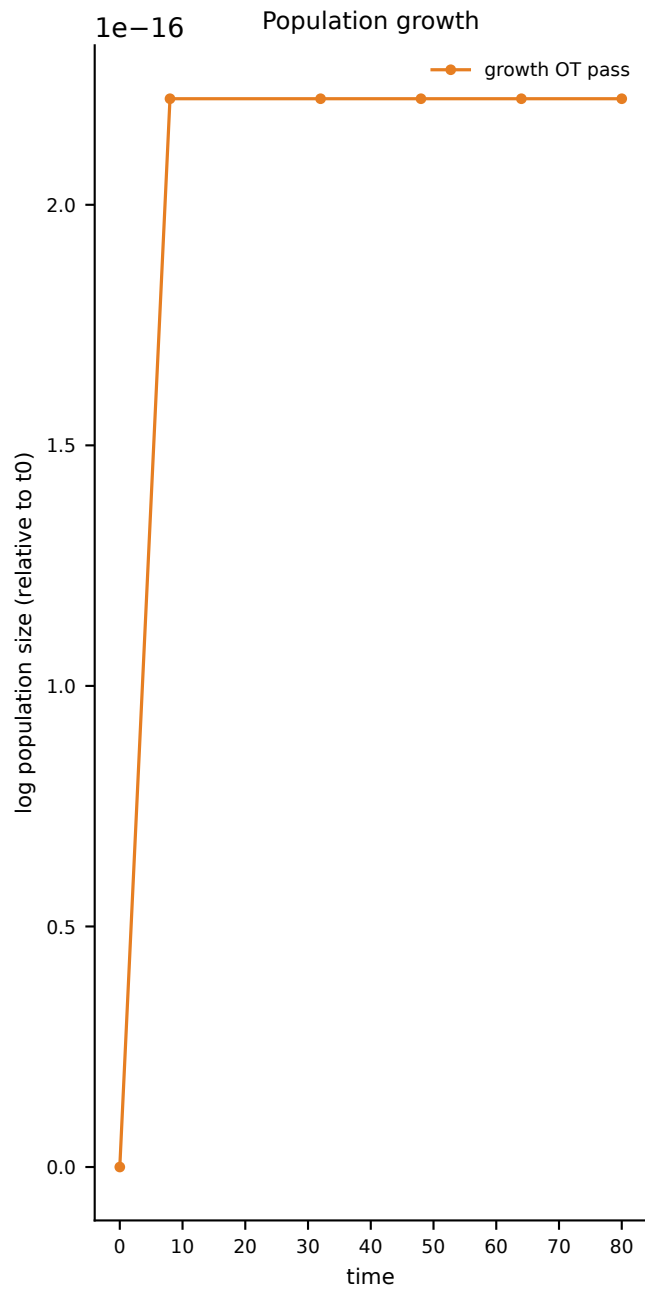


Mean net rate per cell type (cell_type)

Cell type	Anchor	Prior	Learned
precursors	—	—	-0.000994
sympathoblasts	—	—	-0.00901
intermediate	—	—	0.00445
chromaffin	—	—	0.000684

4. Proliferation — population growth and drivers

Relative population size from the mean growth factor per interval; unless Data/population_sizes is given, its absolute level follows the prior.

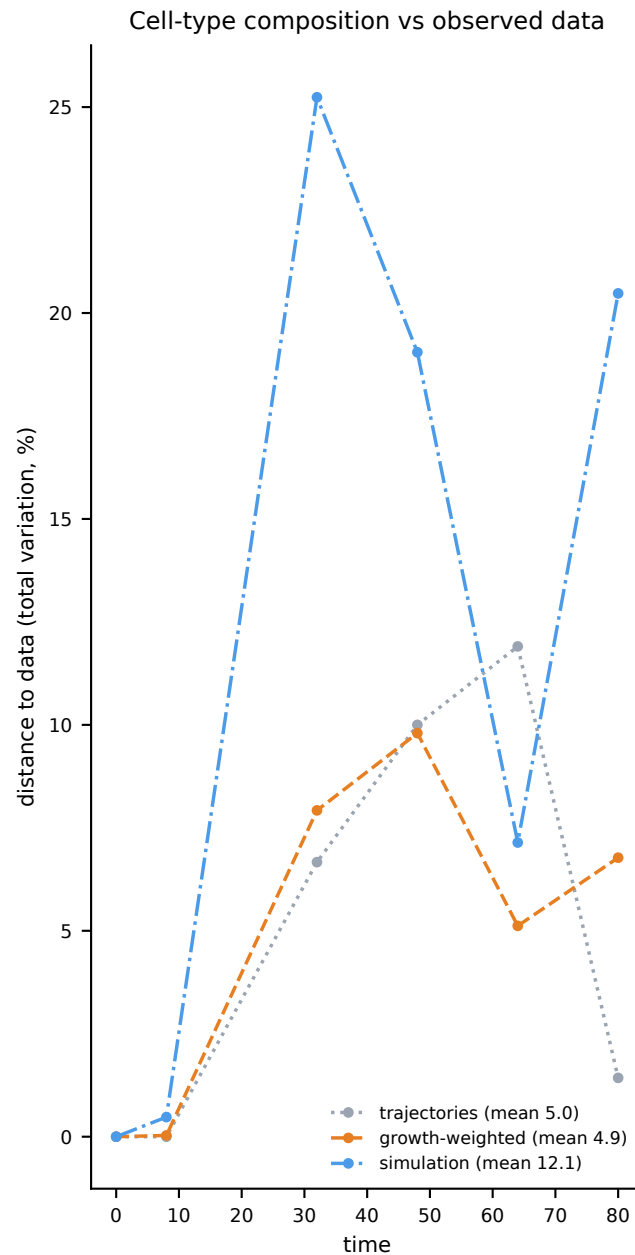
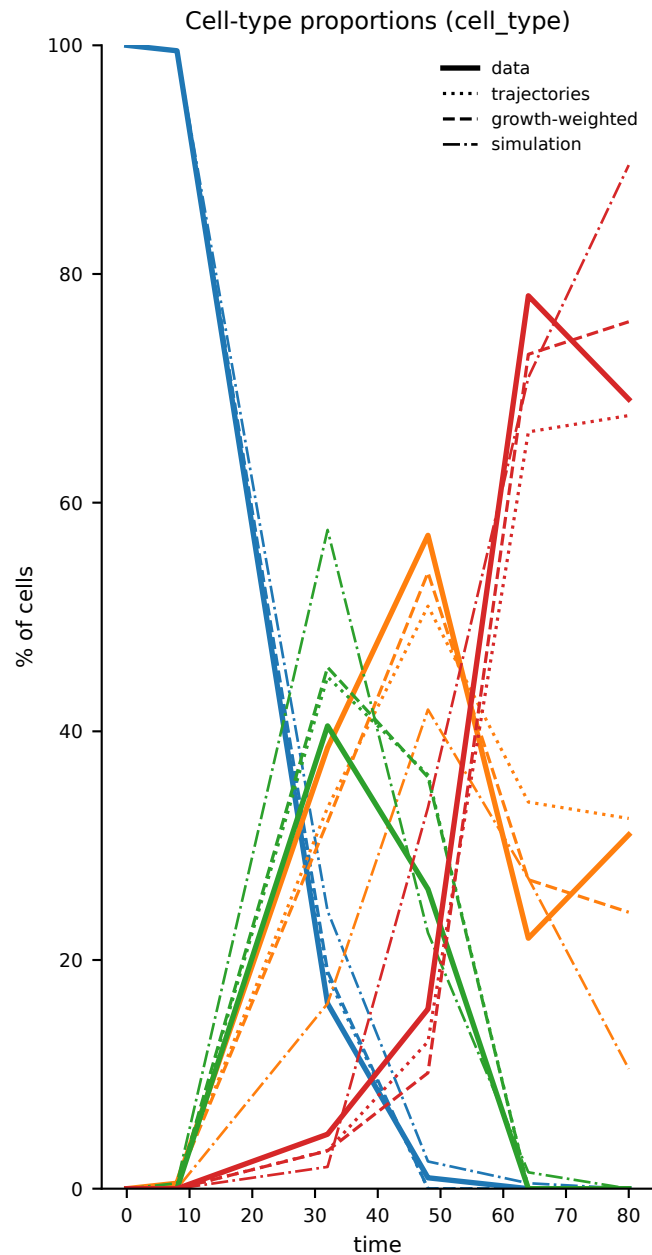


No proliferation MLP
(simulate_with_proliferation = True,
then infer_network_simul)

precursors sympathoblasts intermediate chromaffin

4. Proliferation — growth-weighted trajectories and MLP fit

The OT selection keeps one descendant per ancestor (trajectories without proliferation); weighting each path by its expansion $\exp(JR_{\text{opt}})$ should bring them back to the observed data.

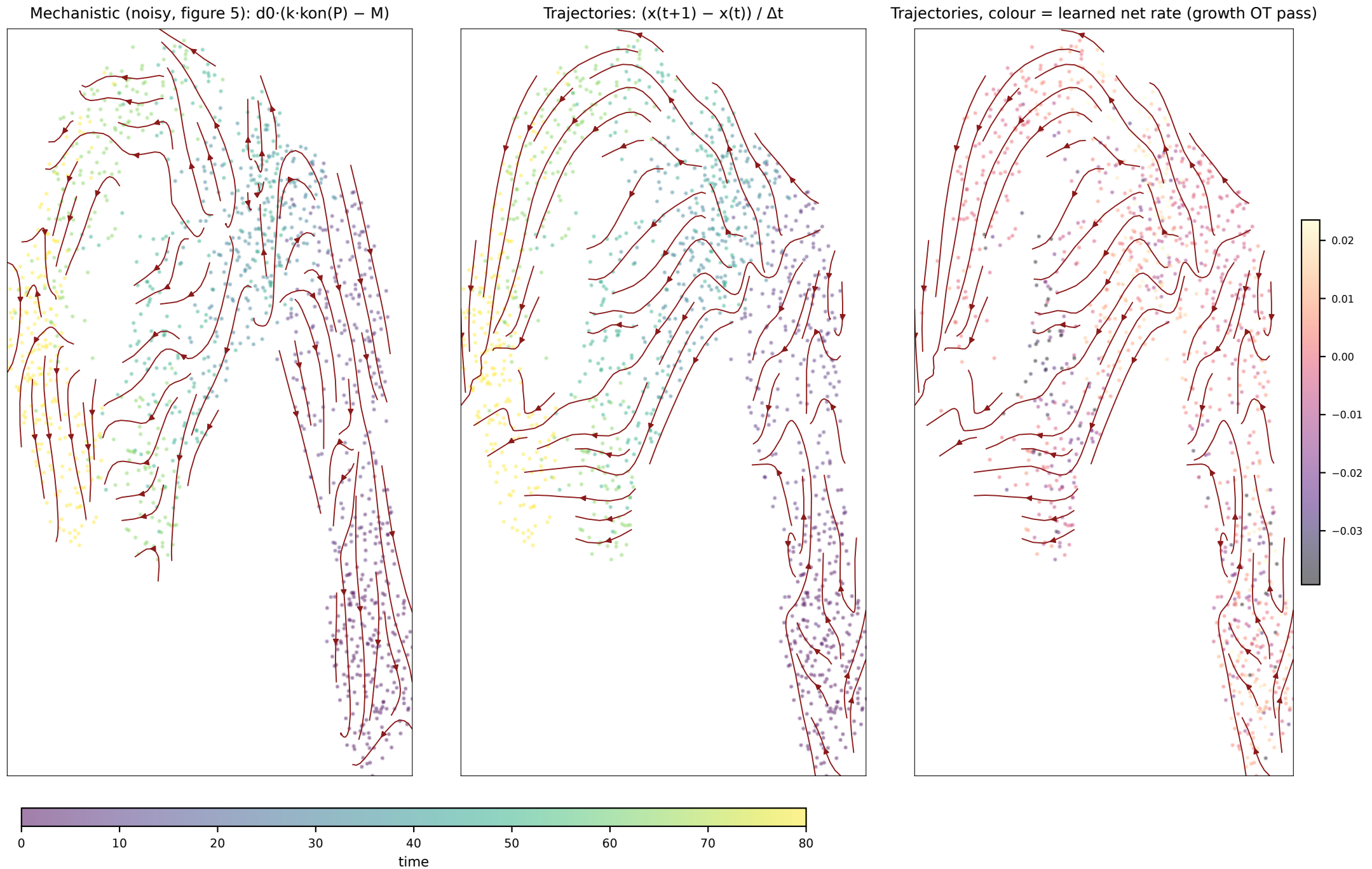


No MLP fit diagnostics
(rerun infer_network_simul
with proliferation)

precursors sympathoblasts intermediate chromaffin

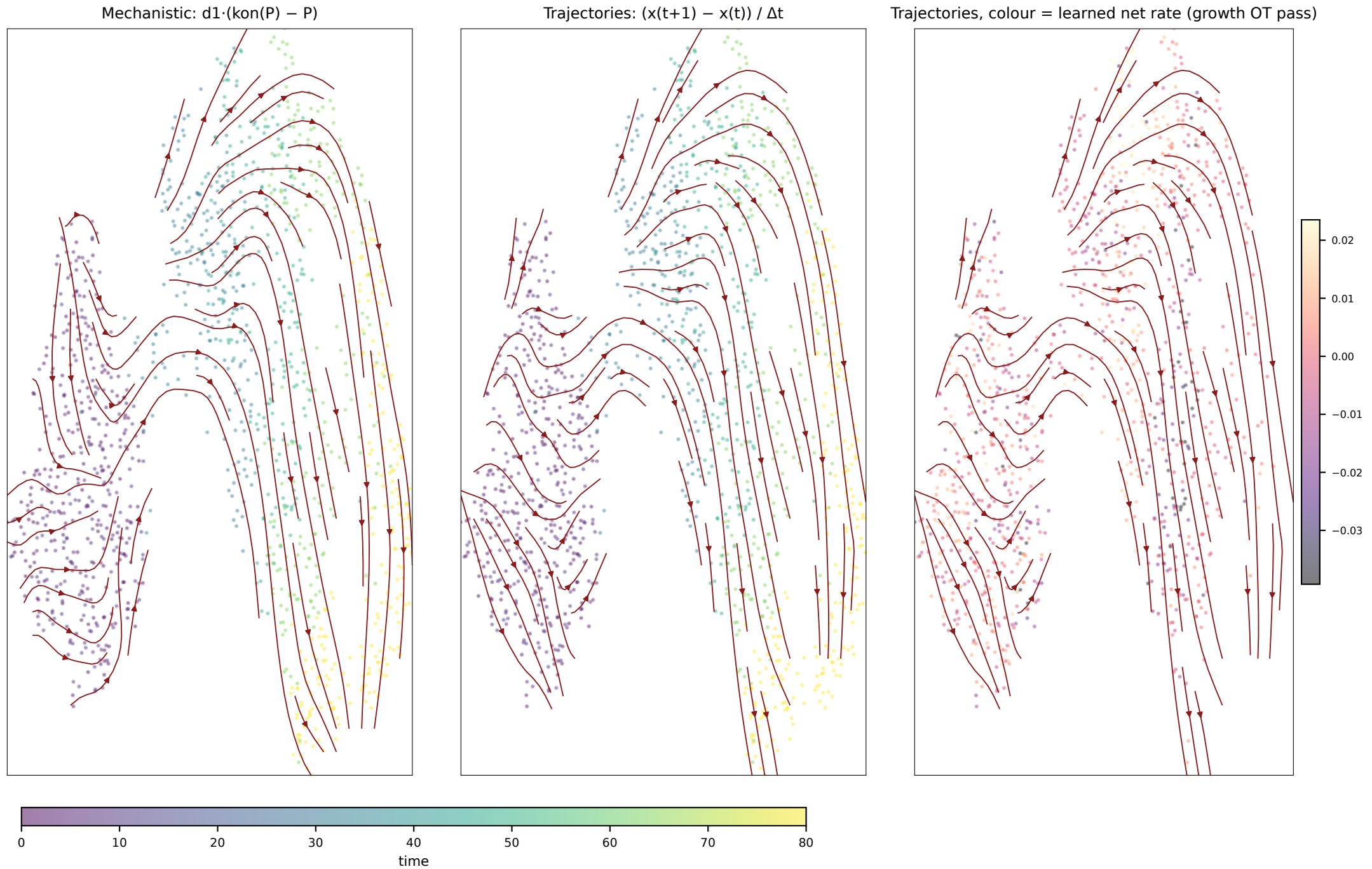
5. Learned dynamics — mRNA velocity fields

1260 trajectory states; kNN-transition projection on a UMAP of log1p NB-sampled mRNA. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.33.



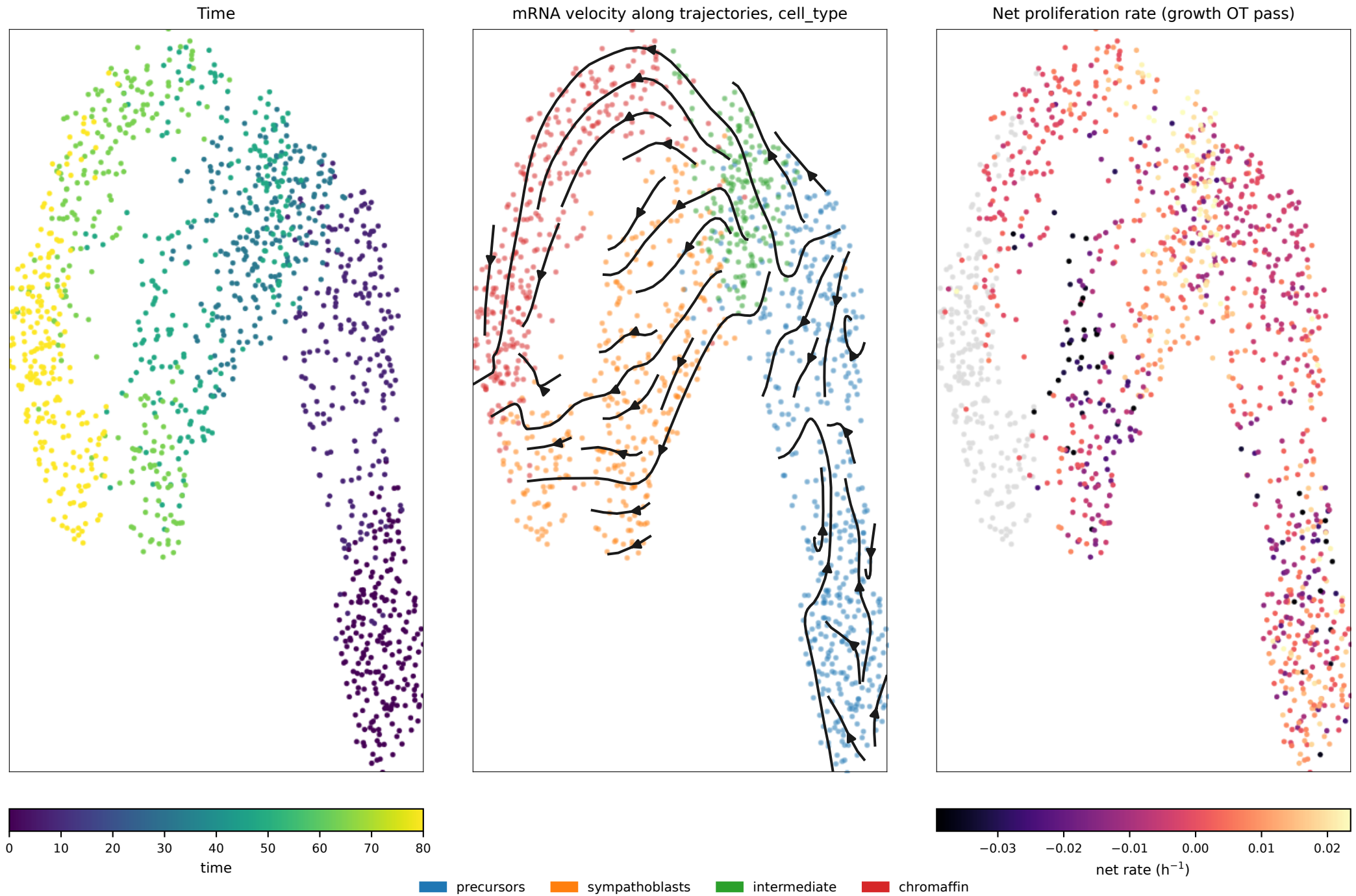
5. Learned dynamics — Proteins velocity fields

1260 trajectory states; kNN-transition projection on a UMAP of protein levels. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.53.



5. Learned dynamics — summary

1260 trajectory states on the UMAP of log1p NB-sampled mRNA; velocity along the inferred trajectories ($(x(t+1) - x(t)) / \Delta t$); background of the velocity field: cell_type.



6. Held-out test cells — predictions with the network fixed

Test cells classified into basins with the training mixtures, trajectories inferred with the training network fixed, then simulated; projected onto the joint UMAP (grey: training).



6. Held-out test cells — quantitative checks

Per gene: 1-D Wasserstein distance between log1p counts, averaged over the common timepoints.

